Sliding Window Complete DSA Notes (Beginner to Advanced)

1. What is Sliding Window?

Sliding Window is an optimization technique used on arrays and strings.

Instead of recalculating a range again and again,
maintain a moving window.

Usually converts:

O(n^2)

into

O(n)

Used for:

- Subarrays
- Substrings
- Maximum/minimum ranges
- Continuous segments

2. Basic Idea

Maintain two pointers:

left

right

Window:

[left .... right]

Expand using right pointer.

Shrink using left pointer.

3. Types of Sliding Window

Two major types:

1. Fixed Size Window

Window length is constant.

Example:

Maximum sum subarray of size K.

2. Variable Size Window

Window grows and shrinks.

Example:

Longest substring without repeating characters.

4. Fixed Window Template

Steps:

1. Add new element.

2. When size reaches k:
calculate answer.

3. Remove left element.

4. Move window.

5. Variable Window Template

Steps:

1. Expand right.

2. Update window state.

3. While condition invalid:
shrink left.

4. Update answer.

Python Fixed Window Example

```
# Maximum sum subarray size k
def maxSum(nums,k):
    window=0
    ans=0

    for right in range(len(nums)):
        window+=nums[right]

        if right>=k-1:
            ans=max(ans,window)

            window-=nums[right-k+1]

    return ans
```

C++ Fixed Window Example

```
int maxSum(vector<int>& nums,int k){
int sum=0;
int ans=0;

for(int r=0;r<nums.size();r++){
sum+=nums[r];

if(r>=k-1){
ans=max(ans,sum);

sum-=nums[r-k+1];
}
}

return ans;
}
```

Java Fixed Window Example

```
int maxSum(int[] nums,int k){
int sum=0;
int ans=0;

for(int r=0;r<nums.length;r++){
```

```
sum+=nums[r];

if(r>=k-1){
ans=Math.max(ans,sum);

sum-=nums[r-k+1];
}
}

return ans;
}
```

Python Variable Window Template

```
def slidingWindow(nums):
    left=0

    for right in range(len(nums)):

        # add nums[right]

        while invalid_condition:

            # remove nums[left]

            left+=1

        # update answer
```

6. Longest Substring Without Repeating Characters

Classic variable sliding window.

LC3

Maintain:

HashSet

If duplicate appears:

remove from left.

Python LC3 Example

```
def lengthOfLongestSubstring(s):
    seen=set()
    left=0
    ans=0

    for right in range(len(s)):

        while s[right] in seen:
            seen.remove(s[left])
            left+=1


        seen.add(s[right])

        ans=max(
            ans,
            right-left+1
        )

    return ans
```

7. Sliding Window + HashMap

Used when frequency matters.

Examples:

- Anagrams
- Minimum Window Substring
- Character Replacement

8. Sliding Window Maximum

Advanced variation.

Use:

Deque

Maintain decreasing order.

Front always stores maximum.

Complexity:

O(n)

9. Common Patterns

Fixed Window:

- Maximum sum size K
- Average size K
- First negative number

Variable Window:

- Longest substring
- Minimum window
- At most K distinct

Advanced:

- Monotonic Queue
- Prefix + Window

Important Leetcode Problems

LC3:
Longest Substring Without Repeating Characters

LC76:
Minimum Window Substring

LC209:
Minimum Size Subarray Sum

LC239:
Sliding Window Maximum

LC424:
Longest Repeating Character Replacement

LC567:
Permutation in String

LC438:
Find All Anagrams

Complexity Cheatsheet

Brute Force:

O(n^2)

Sliding Window:

O(n)

Space:

O(1)

or

O(characters/hashmap size)